GROUP_03 · C1 Report

Who reaches Chile's public hospitals with private insurance? — severity and admission profile of Isapre/Particular vs. FONASA discharges, GRD 2020–2024
Business Intelligence · IIB423T-1 · Universidad del Desarrollo, Facultad de Ingeniería
Team: Miguel Figueroa, Sebastián Bünzli · Group 03 · 29 September 2026

1. The selected problem and how it evolved from W1

1.1 Problem, population, place, period and proposed use

Dimension	Definition
Problem	Little is known about the case mix of privately insured patients who are hospitalized in Chile's public network, and whether they should be monitored as a distinct group.
Population	Hospital discharges (egresos) from public hospitals coded with IR-GRD and published annually by FONASA.
Place	Chile's public hospital network.
Period	Discharge files 2020, 2021, 2022, 2023 and 2024.
Proposed use	Descriptive evidence to support a proposed decision by a public-network management analyst.
Policy objective	Reduce inequities and assure quality of care (Estrategia Nacional de Salud, ENS 2021–2030).

Distinguishing the policy objective from what we measure: the ENS 2021–2030 objective is context for why this problem matters, not a numerical target we directly test. Our indicators — high-severity case share and urgent-admission share — are proxies we selected for this project, not official ENS metrics.

1.2 Documentary support

The Chilean Ministry of Health's *Estrategia Nacional de Salud para los Objetivos Sanitarios al 2030* (Resolución Exenta N°831, issued 11 June 2024; document edition October 2025) lists "reducing health inequities" as one of four Sanitary Objectives for the decade (Ministerio de Salud de Chile, minsal.cl/estrategia-nacional-de-salud-2021-2030/). Separately, a peer-reviewed analysis published in *Salud Problema* (Universidad Autónoma Metropolitana – Xochimilco) argues that, despite successive MINSAL equity strategies, the AUGE/GES mechanism has channelled significant resources toward the private sector, reinforcing a public–private dual model in Chilean health care. This motivated an empirical check of whether that duality is visible in the actual complexity and admission-type data for patients who cross between the two systems.

1.3 Alternatives considered in W1 and why this line was selected

W1 explored two candidates: **Candidate A** (emergency care by age group and facility type, using the SADU/Urgencias dataset) and **Candidate B** (originally framed as a public-vs-private hospital complexity comparison using GRD). Candidate B was refined after W1 once we confirmed that Chile's open GRD data structurally covers only the public network (private hospitals are not required to publish case-mix data) — so the question was reframed from "public vs. private hospitals" to "within the public network, does the case profile differ by the patient's insurance type." Candidate B was selected for C1 because, by the end of W1, it already had richer confirmed fields (relative case weight, severity, insurance type, admission type all native to the same file) and a clearer path to a bounded, answerable question than Candidate A.

1.4 Instructor feedback (W1) and response

Feedback received

> "The main point to improve is effective data inspection: for the first line the structure of the Urgencias file was verified, but the age-by-facility-type analysis was not yet performed; for the second line only one GRD record was inspected, not the full file. In addition, the README was left empty and the notebook was not fully executed, reducing traceability. For C1, the priority is to turn the selected line into a real, reproducible EDA, with results and interpretation grounded directly in the data."

Response in C1:

- Selected Candidate B and loaded the **complete** GRD files for 5 years (2020–2024, ~700K–1M rows/year each) — not a single record.
- Computed 3 KPIs with real values across all 5 years (high-severity share, average case weight, urgent-admission share), by insurance group.
- README completed (member/version/run-order/dependencies/sources); notebook restructured into 8 documented sections and executed top-to-bottom with saved outputs.
- Five Cs checks run directly against the raw files (see Section 3), not asserted without evidence.

2. The seven steps (Class 03)

Step	Evidence
1. Decision	A public-network management analyst deciding whether privately-insured patients treated in the public network should be monitored/managed as a distinct group, versus treating all discharges uniformly regardless of insurance type.
2. Success criteria	A meaningful difference would be a sustained gap in case severity and/or admission type between insurance groups across years, not attributable to a single anomalous year. We do not have a pre-defined numerical target from policy; we treat a gap of several percentage points, repeated across most years, as evidence worth acting on.
3. Business questions	Main: among privately-insured patients who do end up in the public network, is their severity profile lower, similar, or higher than FONASA beneficiaries'? Supporting: does this pattern hold across 2020–2024, and does it coincide with a different admission pathway (scheduled vs. urgent)?
4. Business process	The pathway from patient arrival to hospital discharge and GRD coding. The data captures the discharge record and its case-mix classification; it does not capture the patient's decision process for choosing where to seek care, any prior care sought in the private network, or referral/transfer reasoning beyond the fields present in the file.
5. Entities, events and states	One record = one hospital discharge (egreso) from a public hospital. This is an aggregate clinical event, not a person (the same patient can appear multiple times) and not a facility-level aggregate count.
6. Data representation	Source: FONASA/DEIS annual GRD files, pipe-separated, one row per discharge, ~129 columns (5 used: PREVISION, IR_29301_PESO, IR_29301_SEVERIDAD, IR_29301_MORTALIDAD, TIPO_INGRESO). Coverage: public network only, 2020–2024. No join was performed — the question is fully answered with fields native to GRD alone (see Section 3.4 for the justification of this choice).
7. Measurement	See Section 4 for full formulas, units, numerator/denominator, exclusions and period for each of the 3 KPIs.

3. Data profiling and the Five Cs

3.1 What was inspected

We loaded the complete GRD discharge files for all 5 years directly (not a sample), using the course dictionaries and the files' own headers to confirm field names and meaning before use. Encoding and malformed-line checks (below) were run against the raw files themselves, not assumed from the file extension.

Clean

1 malformed line found out of 932,840 rows (2022 file) — a text field containing a literal "|" character broke the column count for that single line; identified by counting separators per line and safely excluded (0.0001% of the file).

A duplicate check was also run, limited to the 5 loaded columns (no patient identifier was loaded). This returned a very high duplicate count (e.g. 798,322 of 816,909 rows for 2021, ~97.7%). We investigated this rather than reporting it at face value: with only 5 low-cardinality categorical/numeric fields, the vast majority of rows coincidentally match some other row's values by chance — this is a false-positive artifact of the check's design, not evidence of true duplicate patient records. We therefore do not treat this as a real defect, and note that a true record-level duplicate check would require loading a patient identifier, which we did not include in scope.

Consistent

Encoding is **not** consistent across years: 2021 and 2024 are latin1; 2022 and 2023 are UTF-16. Verified by reading each file's header under its assumed encoding and checking that our 5 needed column names were actually readable (a wrong encoding still yields the right column *count* but garbled names, which silently breaks a column-name filter instead of raising an error — this is exactly what happened when we first assumed latin1 for 2023). Also confirmed the same insurance-category labels and TIPO_INGRESO category names are used consistently once encoding is corrected.

Conformed

The 2022 file is published by FONASA as *GRD_PUBLICO_EXTERNO_2022* rather than *GRD_PUBLICO_2022*. Before including it in a multi-year trend, we compared its PREVISION distribution against 2024's: both years are dominated by FONASA (98.0% vs. 98.8%), with PRIVATE roughly double in 2022 (1.61% vs. 0.86%) but still a small minority in both. This does not match the pattern we would expect from a distinct pandemic-era private-referral cohort (which would show a much larger PRIVATE share), so we concluded "EXTERNO" most likely reflects a naming choice for that year's publication rather than a structurally different population, and included it in the trend — while flagging the near-doubling of the PRIVATE share as a residual observation worth noting, not resolving further.

Current

Data covers the 5 most recent years for which FONASA has published complete annual GRD files (2020–2024) at the time of this project. This window is appropriate for the proposed decision (an ongoing monitoring question), and additionally lets us observe a pre-pandemic year (2020) alongside pandemic and post-pandemic years.

Comprehensive

We worked from the complete discharge files for each year (roughly 700K to 1.4M rows per year before filtering), not a subset of hospitals or regions. Within each file we deliberately scoped to the 5 fields relevant to this question, out of ~129 available columns (diagnosis codes, procedure codes, transfer dates, newborn fields, etc. were left out as out of scope for this line, not because they were inaccessible). What remains unobserved: patient-level identifiers (not loaded, limiting the duplicate check as noted above), and any private-network activity for these same patients, which is not published in open data.

3.4 Why no join was performed

A join with the Establecimientos dataset (proposed in W1, via COD_HOSPITAL, and left explicitly marked as "proposed, not tested") was reconsidered for C1 and **not pursued**. The refined question is fully answerable with

fields native to the GRD file alone (PREVISION, severity, admission type), so a join would not change the answer to the central question and risked being a forced merge without analytical payoff. This remains a documented possible extension (e.g. to add hospital-level attributes such as region or level of care) rather than a requirement we skipped.

4. Measurement: how the KPIs are calculated

All three KPIs are computed per insurance group (FONASA vs. PRIVATE) and per year, over the **valid** discharge population: records with a mappable insurance category, excluding ambulatory-surgery cases with no real clinical grouping (IR_29301_SEVERIDAD and IR_29301_MORTALIDAD both "0", confirmed in the 2024 file to be 212,041 of 212,051 such cases – 99.995% – same-day ambulatory surgery, where these clinical scales do not apply), and excluding severity coded "DESCONOCIDO".

KPI	Formula	Notes
KPI 1 % high-severity cases	n (SEVERIDAD = 3) / n (valid discharges) × 100	Level 3 is the highest severity level the GRD grouper assigns — not a cutoff we chose.
KPI 2 Average GRD weight	Σ PESO / n (valid discharges)	PESO parsed from text (comma decimal → float). This is a *relative* case-mix weight, not a cost figure in Chilean pesos.
KPI 3 % urgent admission	n (TIPO_INGRESO = URGENCIA) / n (valid discharges) × 100	Category name verified against the real data (value_counts) before filtering, in both 2020 and 2024, rather than assumed.

On the question of projecting government costs: *the working question slide for this project asks whether this pattern "could be projected for government costs." With the current data,* ***this cannot be done****: IR_29301_PESO is a relative case-mix weight, not a monetary value, and no cost or reimbursement field is present in the public GRD file. A cost projection would require either FONASA's GRD payment-value tables (which convert weight to a monetary base rate, not published in this open dataset) or a different data source entirely. We flag this explicitly as an unanswered question and a concrete data gap for any future extension of this line, rather than presenting an estimate we cannot support.*

5. EDA results and interpretation

5.1 Severity gap by insurance group

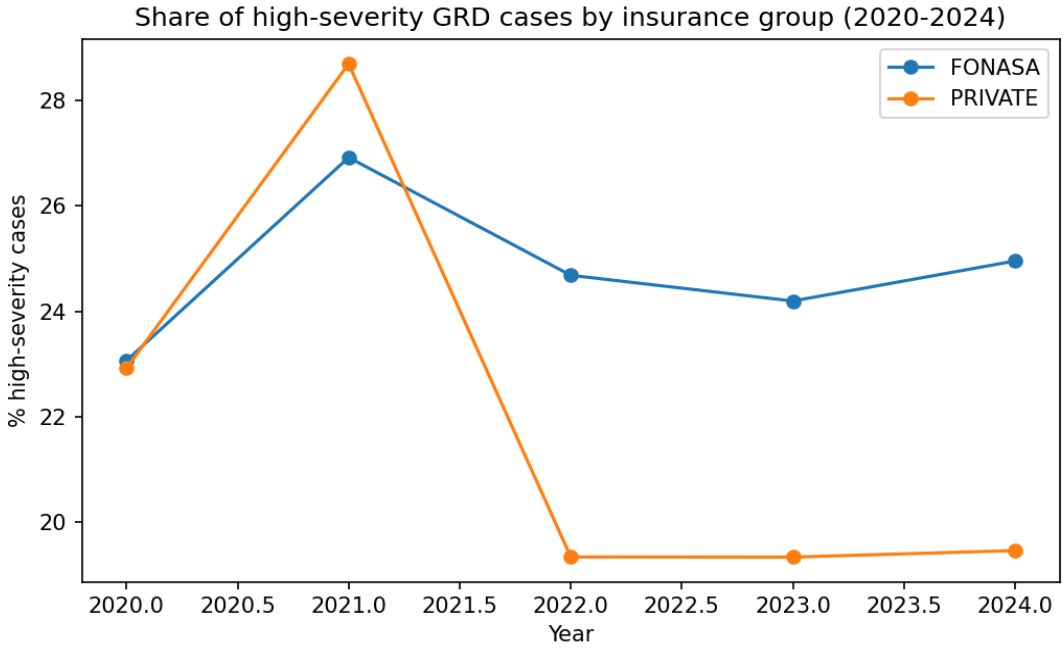


Figure 1. Share of high-severity GRD cases (KPI 1) by insurance group, 2020–2024.

FONASA and PRIVATE are nearly equal in 2020 (~23.1% vs. ~22.9%). Both spike sharply in 2021 (FONASA 26.92%, PRIVATE 28.70% — the only year where PRIVATE exceeds FONASA). From 2022 onward, PRIVATE drops and stabilizes at 19.3–19.5%, while FONASA remains higher at 24.2–24.95%, a gap of 4.9 to 5.5 percentage points sustained across three consecutive years.

The 2021 anomaly

2021 is treated as an isolated anomaly rather than a sign that the 2022–2024 pattern is unstable. Supporting evidence: the count of ambulatory-surgery cases with no clinical grouping assigned was markedly lower in 2021 (102,938) than in other years (150,000–212,000), and 2020 — a genuine pre-pandemic baseline — shows neither the 2021 spike nor the sustained 2022–2024 gap, but a near-zero difference of its own. This is consistent with fewer elective/ambulatory procedures being performed during the peak pandemic year, which changes the mix of cases entering the severity calculation for that year specifically, rather than reflecting a genuine two-year shift in how privately-insured patients use the public network. *Open observation, not further investigated given time constraints:* 2021 also shows an unusually high count of unmapped PREVISION categories (11,601 vs. ~24–59 in other years).

5.2 Urgent admission share by insurance group

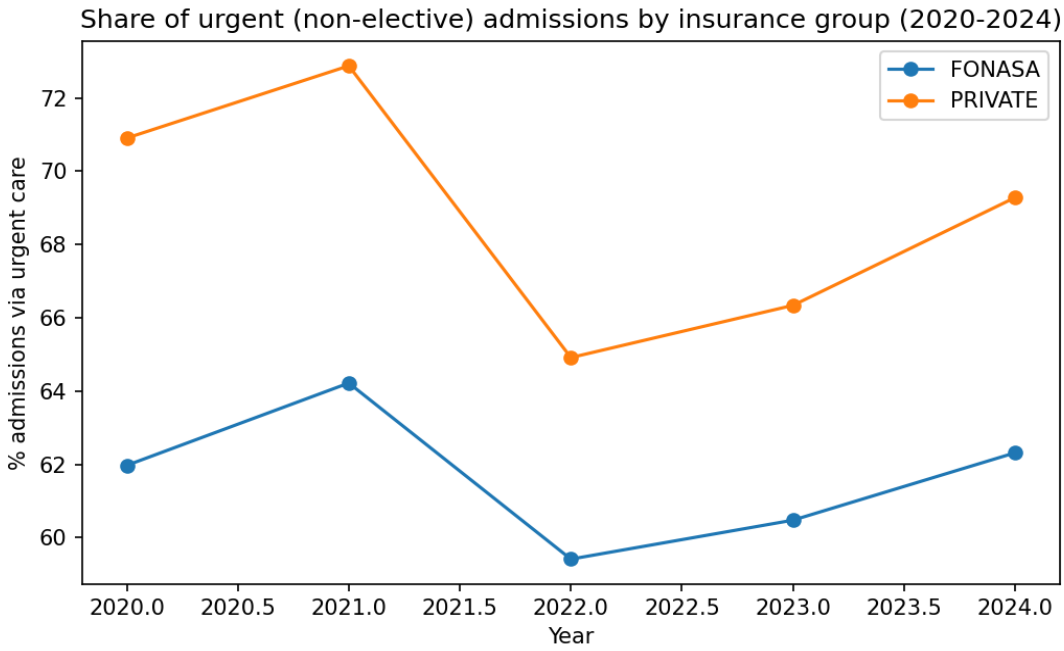


Figure 2. Share of admissions via urgent (non-elective) care (KPI 3) by insurance group, 2020–2024.

PRIVATE is consistently higher than FONASA in **all 5 years**, including 2020 and 2021 (e.g. 2020: 70.9% vs. 62.0%; 2024: 69.3% vs. 62.3%). Unlike the severity KPI, this pattern is not affected by whatever produced the 2021 severity anomaly — it is the more robust of the two findings in this line.

5.3 Working thesis

> **Working thesis (interpretation, not a tested causal claim)**
>
> Privately-insured patients appear to treat the public network as a fallback for minor, unplanned incidents — not a substitute for their own care. They almost never enter through scheduled/elective admission, but the cases they do bring in tend to be less clinically severe than FONASA's — suggesting that when something is genuinely serious, it is more likely resolved within their own private network instead of showing up in this data at all.

5.4 What this analysis can and cannot establish

Within the evidence:

Privately-insured patients treated in the public network show lower severity and higher urgent-admission rates than FONASA beneficiaries across most of the study period (2020, 2022–2024 for severity; all 5 years for admission type).

Cannot conclude:

- A full public-vs-private *system* comparison — private-hospital GRD data is not publicly available, so we only observe the subsample of privately-insured patients who do reach the public network, not those who never do.
- The working thesis above as established fact — it is our interpretation of two complementary patterns, not a tested causal claim; we cannot observe what happens in the private network to confirm the counterfactual.
- Any monetary/cost figure — see Section 4 on why a government-cost projection is not currently feasible with this data.

5.5 Future ML / dashboard potential

ML task: predict/classify public hospitals at risk of a rising high-severity case load. Unit of analysis: hospital × month. Inputs available at decision time: historical PESO/SEVERIDAD distribution, insurance mix, admission-type mix. To check before further work: sample size per hospital-month, and whether Establecimientos-level attributes (region, level of care) add predictive value.

Dashboard: proposed user is a FONASA public-network management analyst, to support case-load balance and risk-compensation discussions between subsystems. Core indicators: case-mix weight and admission-type split by insurance group, tracked over time and by hospital — directly grounded in the EDA findings above.

Process Record

Decisions and checks

Date/stage	Decision	Reason / evidence
24–25 Sep (W1)	**Selected 4 of 5 course sources; excluded cartografía_censo_2024**	Not central to either candidate question — Reviewed both candidate questions; neither needs per-capita normalisation for a first cut
25 Sep (W1→C1)	**Reframed Candidate B from "public vs. private hospitals" to "within the public network, by insurance type"**	Verified GRD is structurally a public-network payment mechanism (FONASA institutional presentation, Aug. 2024); private-provider GRD agreements are punctual, not systematic (La Tercera, 2021)
Sep (C1)	**Corrected assumed latin1 encoding to UTF-16 for 2022 and 2023**	Column-name validation (usecols) failed silently/loudly at load time — Verified via raw byte/header inspection before trusting any KPI from those years
Sep (C1)	**Investigated a 97.7% "duplicate rows" result instead of reporting it directly**	Result looked implausibly high for a real data defect — Diagnosed as a false-positive artifact of checking only 5 low-cardinality columns with no patient ID; documented as a check limitation, not a data defect
Sep (C1)	**Added 2020 as a pre-pandemic baseline year (addendum notebook)**	Whether the 2021 severity anomaly was a one-year event or a two-year pandemic pattern — Compared 2020's severity gap against 2021 and 2022–2024; concluded 2021 is the isolated year, not part of a broader pandemic pattern (Section 5.1)
Sep (C1)	**Decided not to pursue the Establecimientos join proposed in W1**	Whether a join was necessary to answer the refined question — Confirmed the question is fully answerable with native GRD fields alone; documented as a deliberate choice, not an omission

Instructor feedback and response

See Section 1.4 for the feedback received after W1 and the corresponding response. No additional instructor feedback had been received as of this C1 submission.

Individual contributions

Member	Contribution	Evidence location
Sebastián Bünzli	Data loading and cleaning pipeline (encoding fixes, Five Cs checks, KPI calculation), 2020 addendum notebook	analysis/01_grd_severity_analysis.ipynb, analysis/02_grd_2020_addendum.ipynb
Miguel Figueroa	Presentation slides; data-cleaning support; contributions to this report	presentation/GROUP_03_C1_Slides.pdf; report/GROUP_03_C1_Report.pdf

AI use and verification

Tool: Claude (Anthropic). **Purpose:** troubleshooting local Python/PowerShell/encoding errors while loading course data; exploring data dictionaries and raw file structure; drafting and debugging analysis code; organizing the README, this report, and the presentation slides.
What we verified: every documentary source cited in Section 1.2 was independently retrieved and read at its original URL before being cited. Every data finding reported (encodings, malformed-line counts, KPI values, the duplicate-check artifact) comes from code executed by the team against the real files, not from AI-generated figures. The AI's suggestion to treat the 97.7% duplicate result at face value was not made; instead we asked for and

confirmed the cause before writing it up (Section 3, "Clean").
What we changed or rejected: an earlier presentation draft (built from a generic template) was discarded in full except for the title and question slides; formulas and Five-Cs wording were revised for accuracy (e.g. correcting "last 6 years" to the actual 5-year window, 2020–2024) before inclusion in this report.

Individual reflection

Sebastián Bünzli: Contribution: data loading and cleaning pipeline across all 5 years, including diagnosing the encoding and duplicate-check issues documented in Section 3. Learning gained: how to extract, clean and prepare real public-sector data for analysis — encoding issues, malformed records, and category inconsistencies that don't show up until you actually try to load the file. Unresolved uncertainty: I would like to be able to project the fiscal spending programmed for these hospitals and its growth over time, as a realistic extension given what this analysis has already found — but this is not something the current GRD data supports (see Section 4 on why a cost projection is not currently feasible), so it remains an open question for future work rather than something we resolved here.

Miguel Figueroa: Contribution: presentation slides, support on data cleaning, and contributions to this report. Learning gained: polished my knowledge of data analysis, including how to use the seven steps (Class 03) to make analytical reasoning easy to locate and communicate. Unresolved uncertainty: it would be interesting to understand *why* privately-insured patients resort to the public network — convenience at certain times, a perception of better care for specific conditions, or an underlying economic reason. Interviewing these patients could shed light on this, though the assignment does not require and we did not conduct interviews, so this remains an interpretation we cannot confirm with the current data (see Section 5.4).

References

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- FONASA. GRD discharge files (SA_21_V1.2 layout), 2020–2024, course-provided extract. https://datosabiertos.fonasa.cl